



Peters the Great Saint-Petersburg polytechnic university
Institute of cybersecurity and computer science
High school of artificial intelligence technologies
02.04.01 Math and computer science

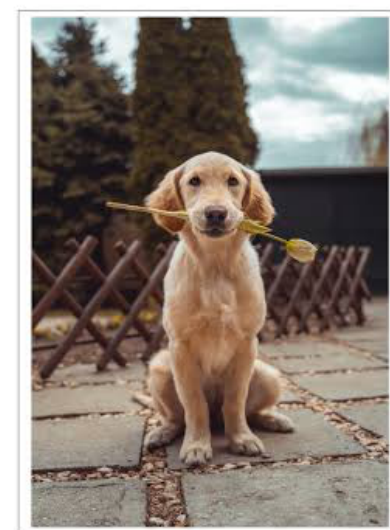
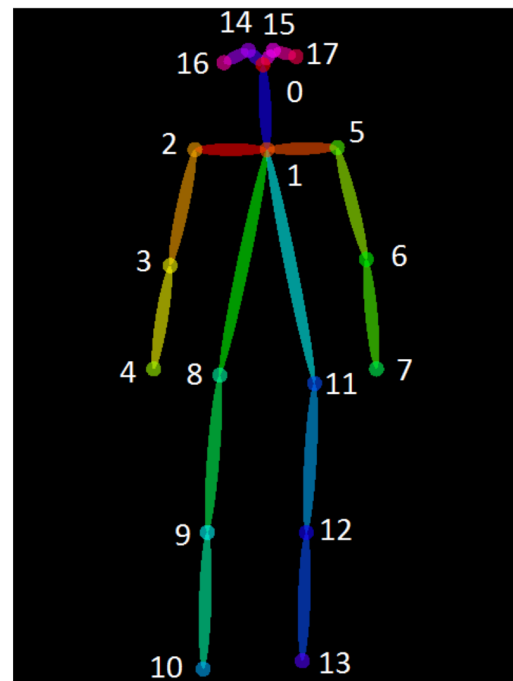
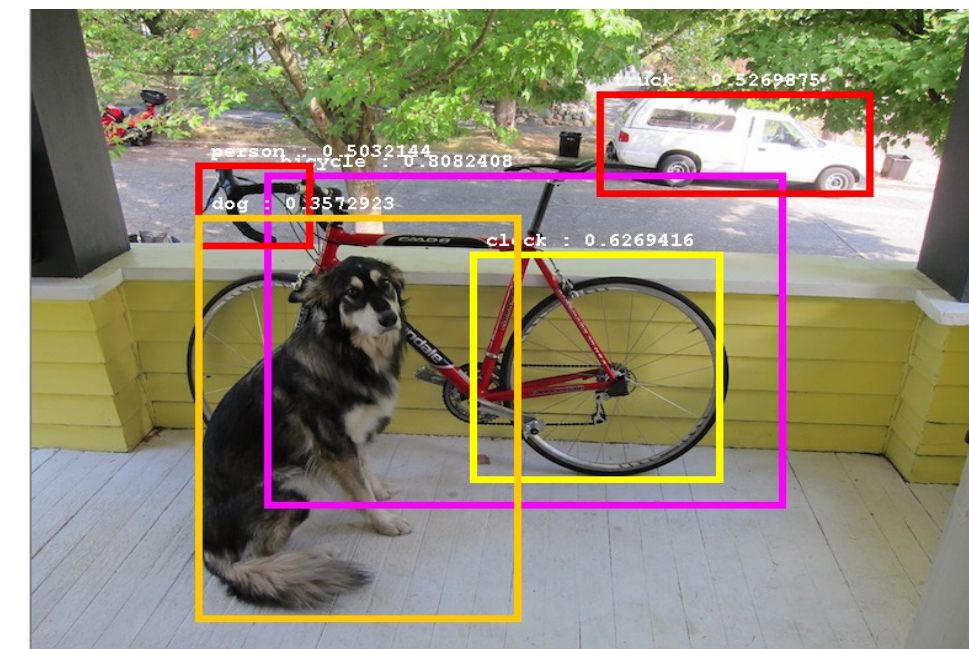
Instance and semantic based segmentation models

Introduction

- **Is there a segmentation model capable of solving all segmentation tasks?**
(*no...*)
- **Problems of modern segmentation:**
 - Slow inference
 - Low quality on tiny or small objects (based on mIoU and mDice metrics)
 - ViT using all pixels
 - Quality is worse after quantization (down to 4 / 8 bytes)

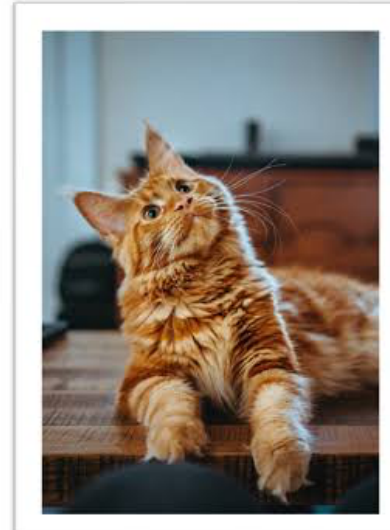
Computer vision

Computer vision - field of AI uses deep learning and neural networks to recognize objects, people, and patterns with high degrees of accuracy. In other words, it replicates human sight and the cognitive ability to interpret visual data.



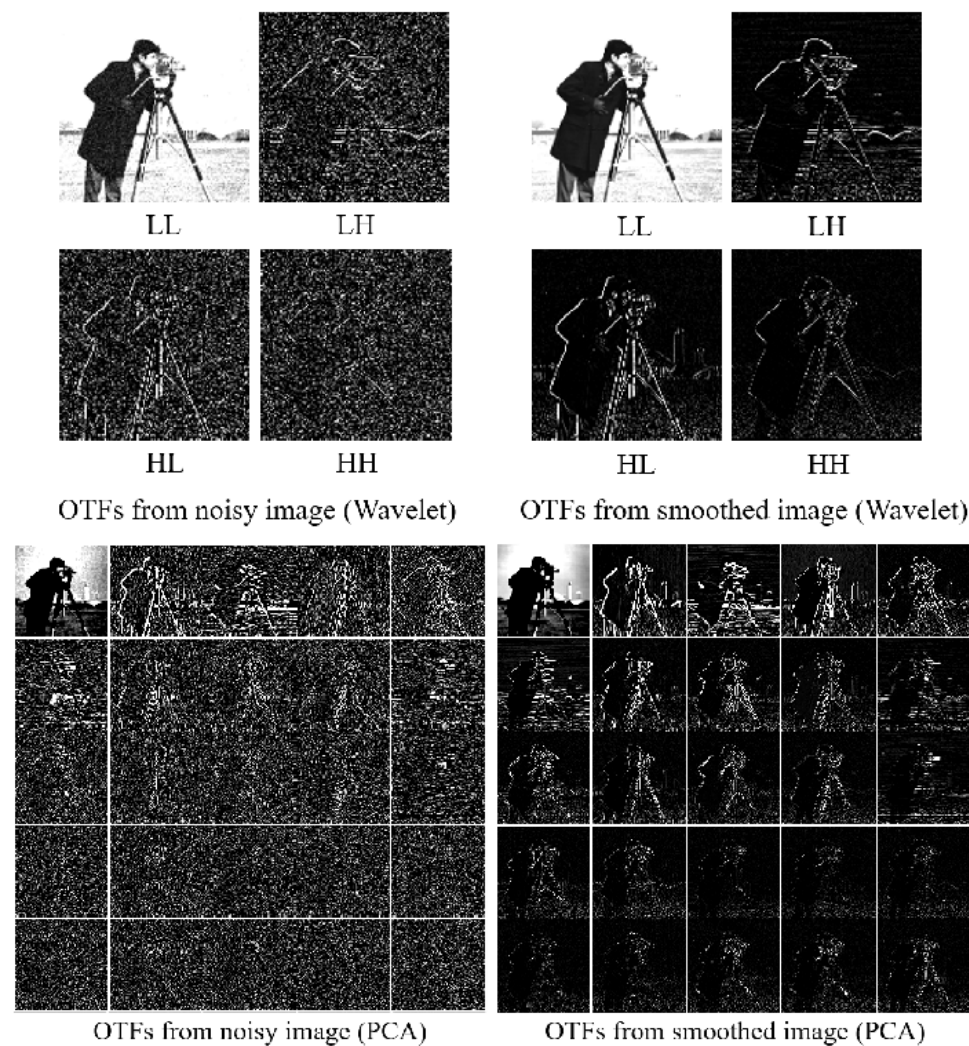
Dog

VS



Cat

Microsoft Azure

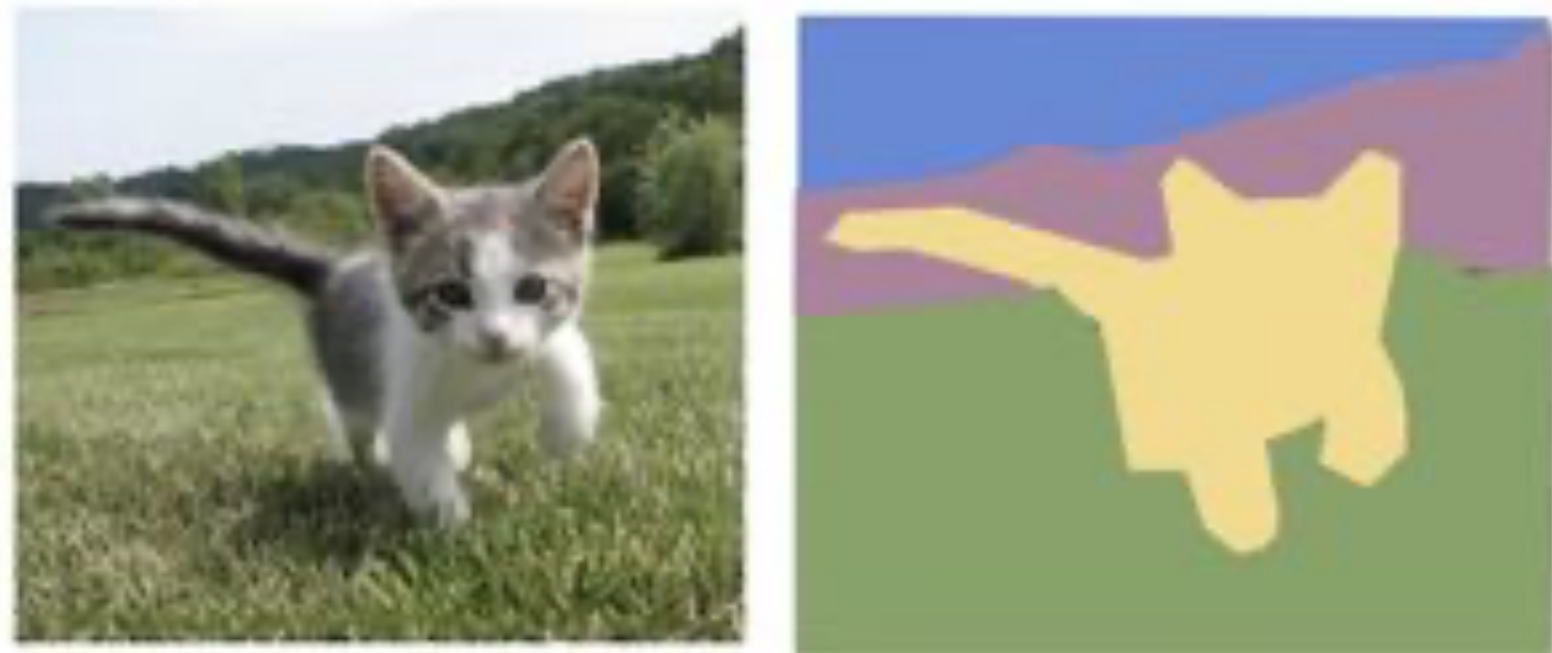


$$I(H, W, C) \rightarrow T_r,$$

T_r - semantic space, h - image height, w - image width

What is segmentation

- Object segmentation - image splitting into semantic regions or instance of objects.



$$F : I(H, W, C) \rightarrow 1, 2, \dots, Q,$$

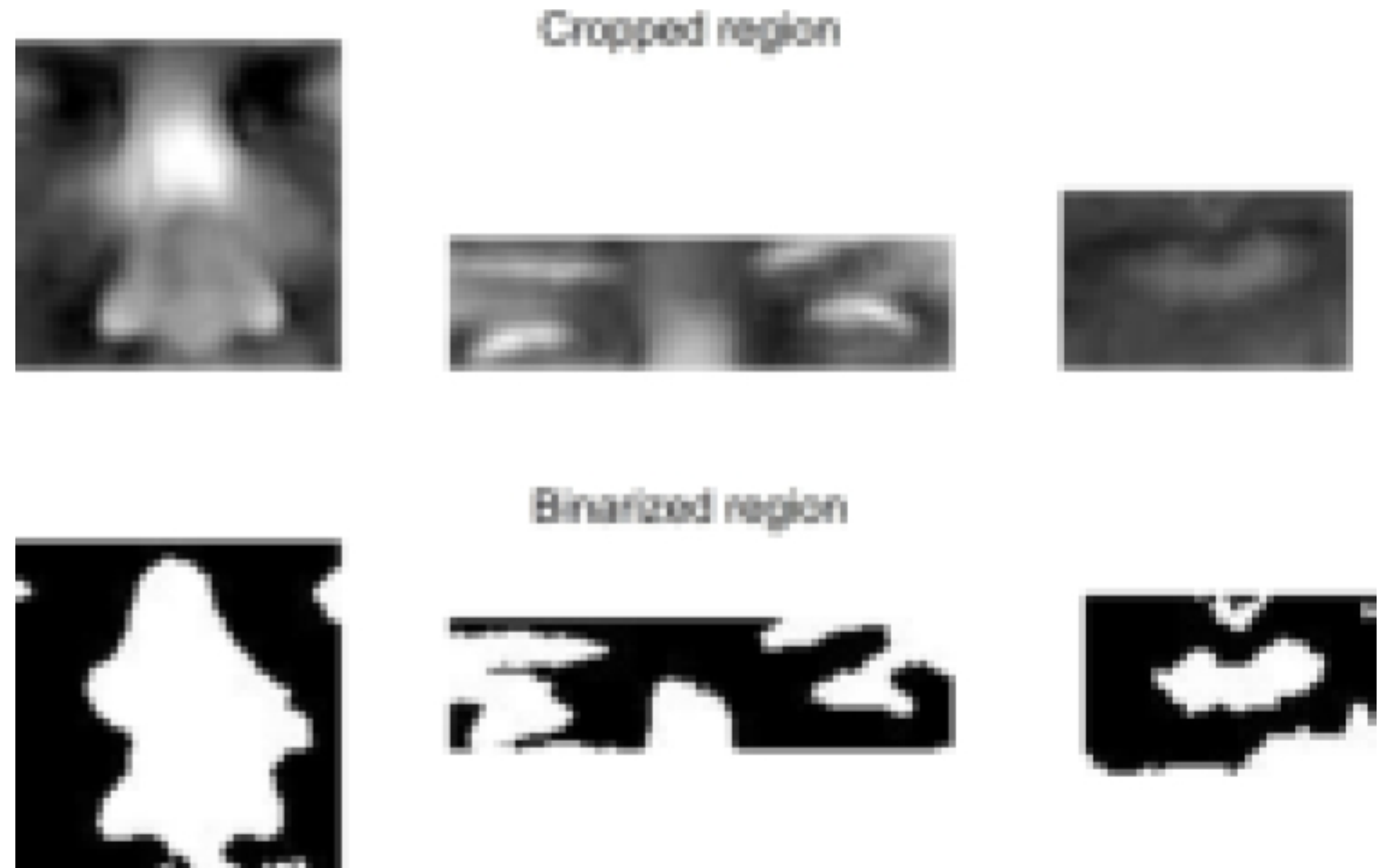
Q - numbers of segments

Types of segmentation:

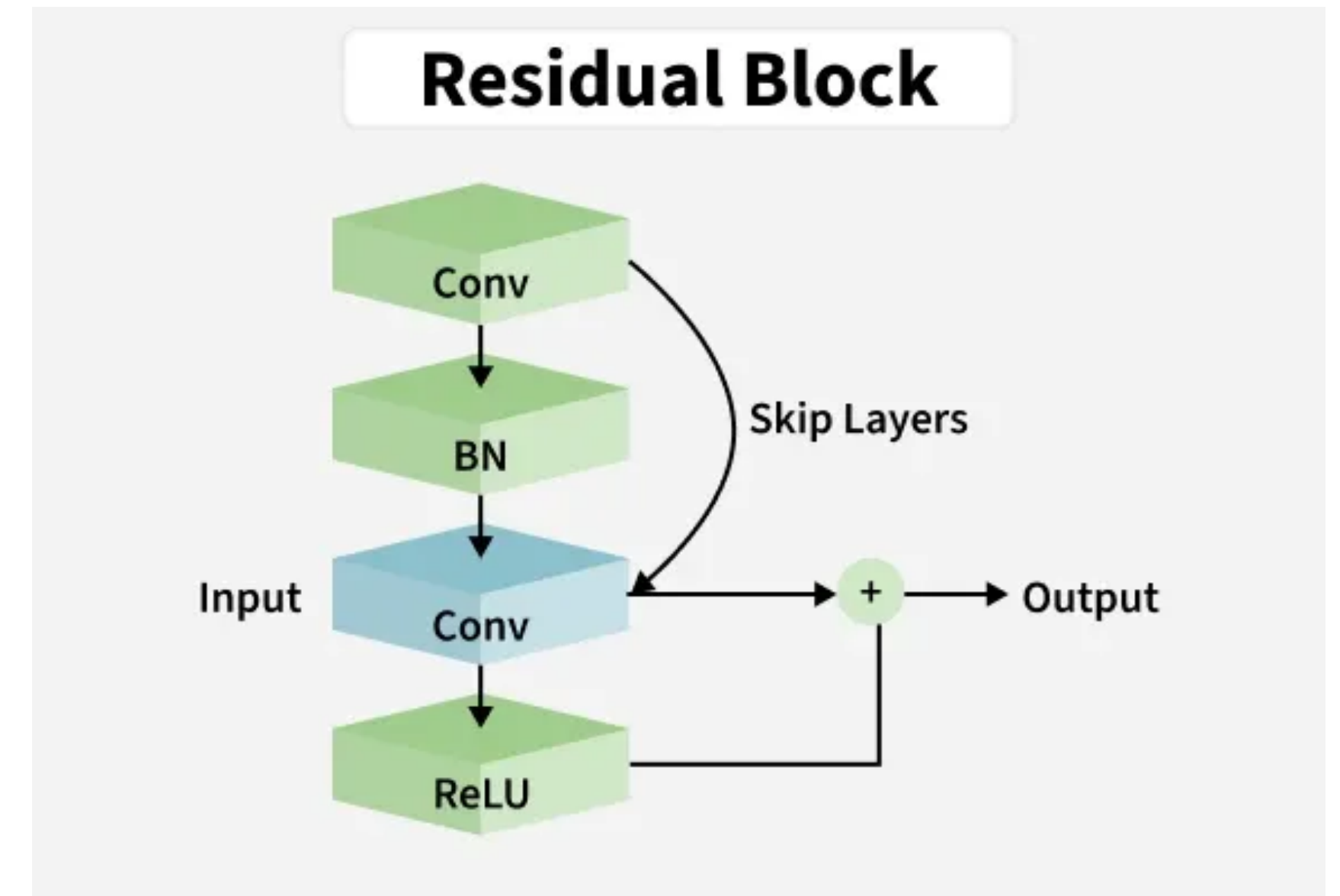
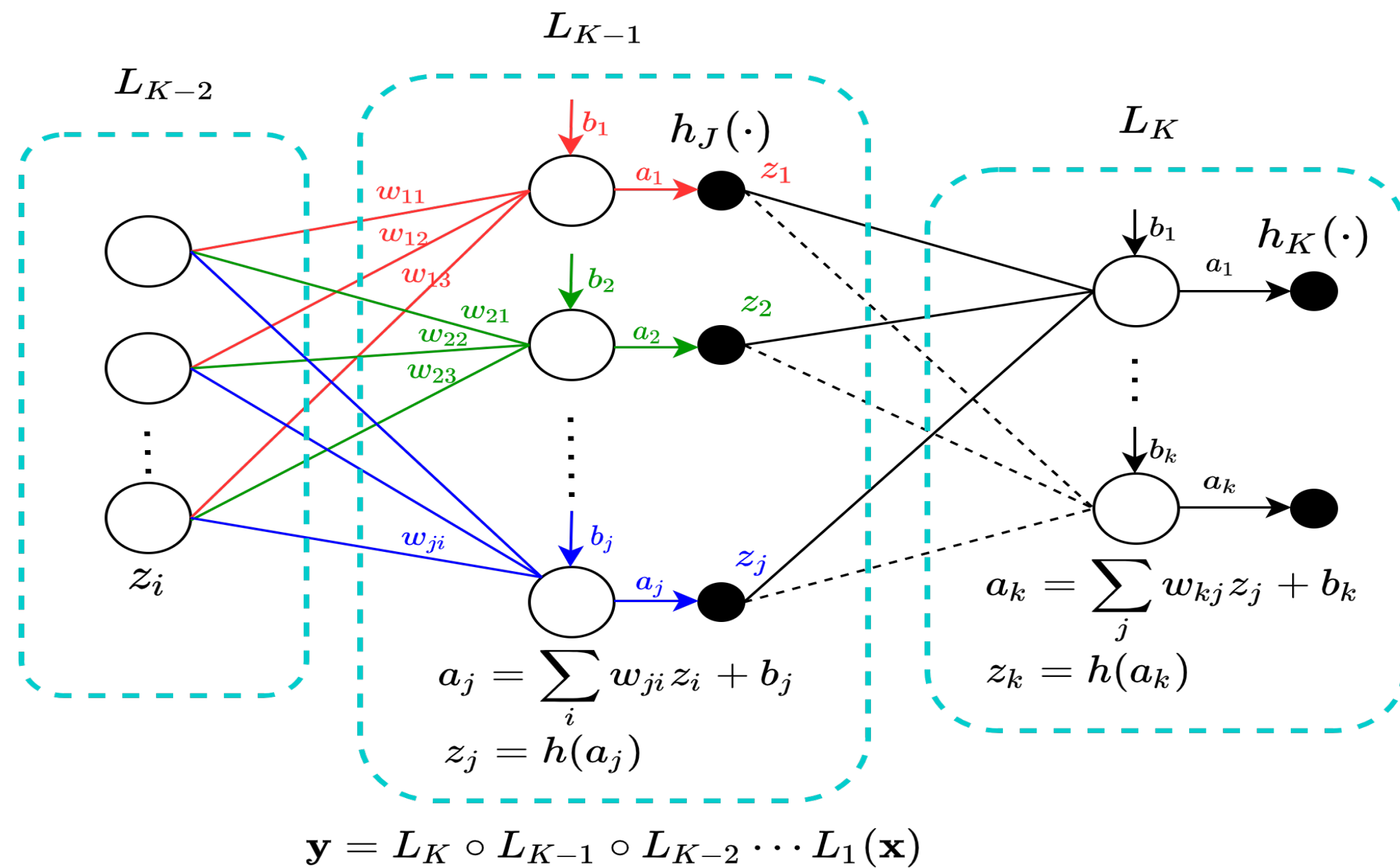
- Semantic segmentation
- Instance segmentation
- Algorithmic segmentation (images with $(H, W, 2)$ shape)

Algorithmic-based segmentation

- Algorithmic segmentation
 - Gray-level segmentation
 - Haar filter
 - Pixel intensity
- Problems
 - High-quality images
 - Better for 2 channels images
 - The majority of segmentation tasks are algorithmically unsolvable



Deep Learning

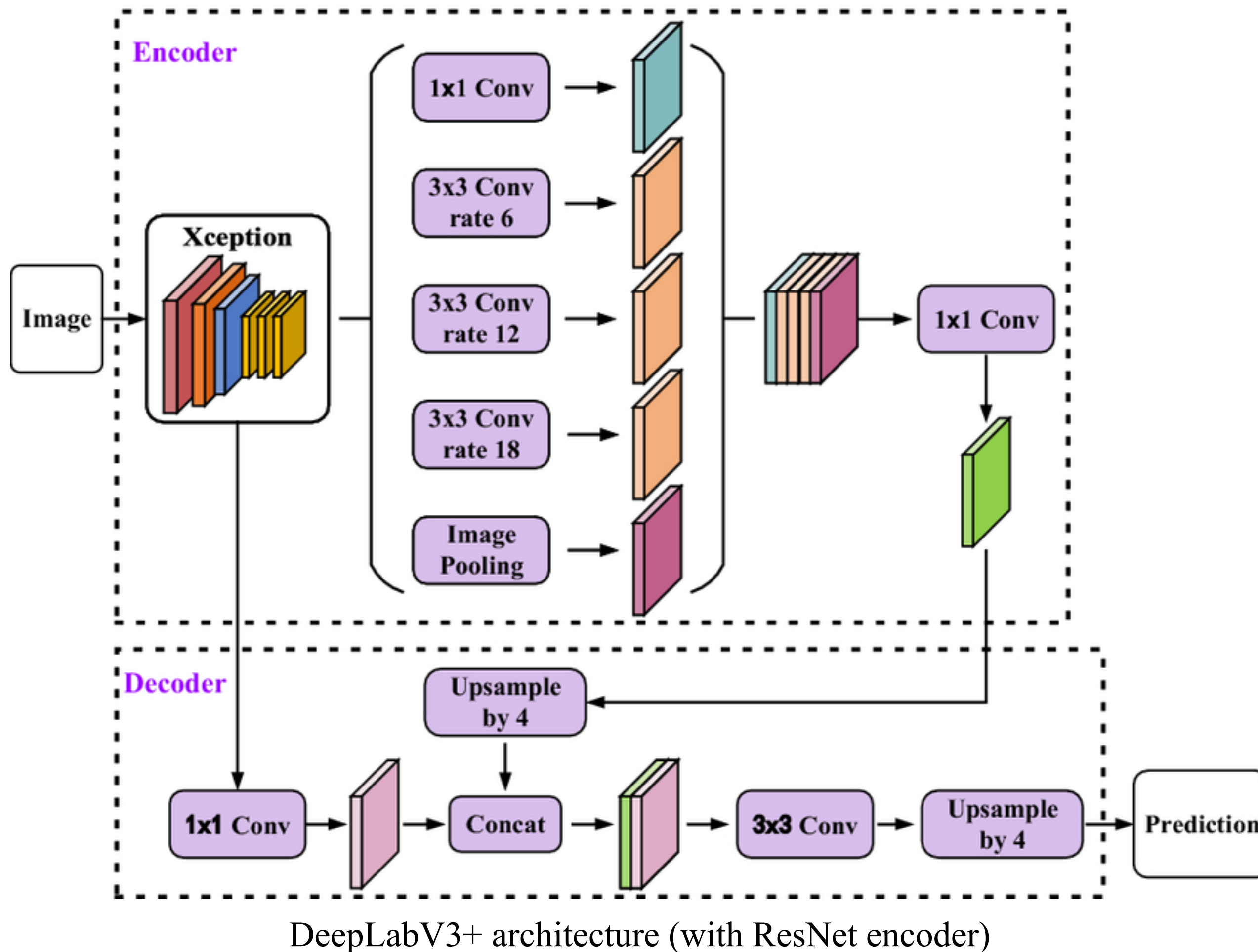


- Classic neural networks
 - Gradient vanishing
- Deep learning (thanks for ResNet!)
 - More than 100 layers NN

$$L_K(x) = F(x) + x$$

$$L_{K_m}(x) = \frac{dF(x)}{dx} + \frac{dx}{dx} = \frac{dF(x)}{dx} + 1$$

Atrous convolution based neural networks



- Residual Network ResNet
(training on residuals) : (BN → ReLU)

$$H(x) = \sigma(F(x, \{W_i\}) + W_s x), W_s : linear_{proj}$$

x - input tensor, $F(x, \{W_i\})$ - residuals, $\{W_i\}$ - set of weights, W_s - convolution 1x1

- Atrous convolution (feature extraction)

$$y[i] = \sum_{k=1}^K x[i + r \cdot k] \cdot w[k]$$

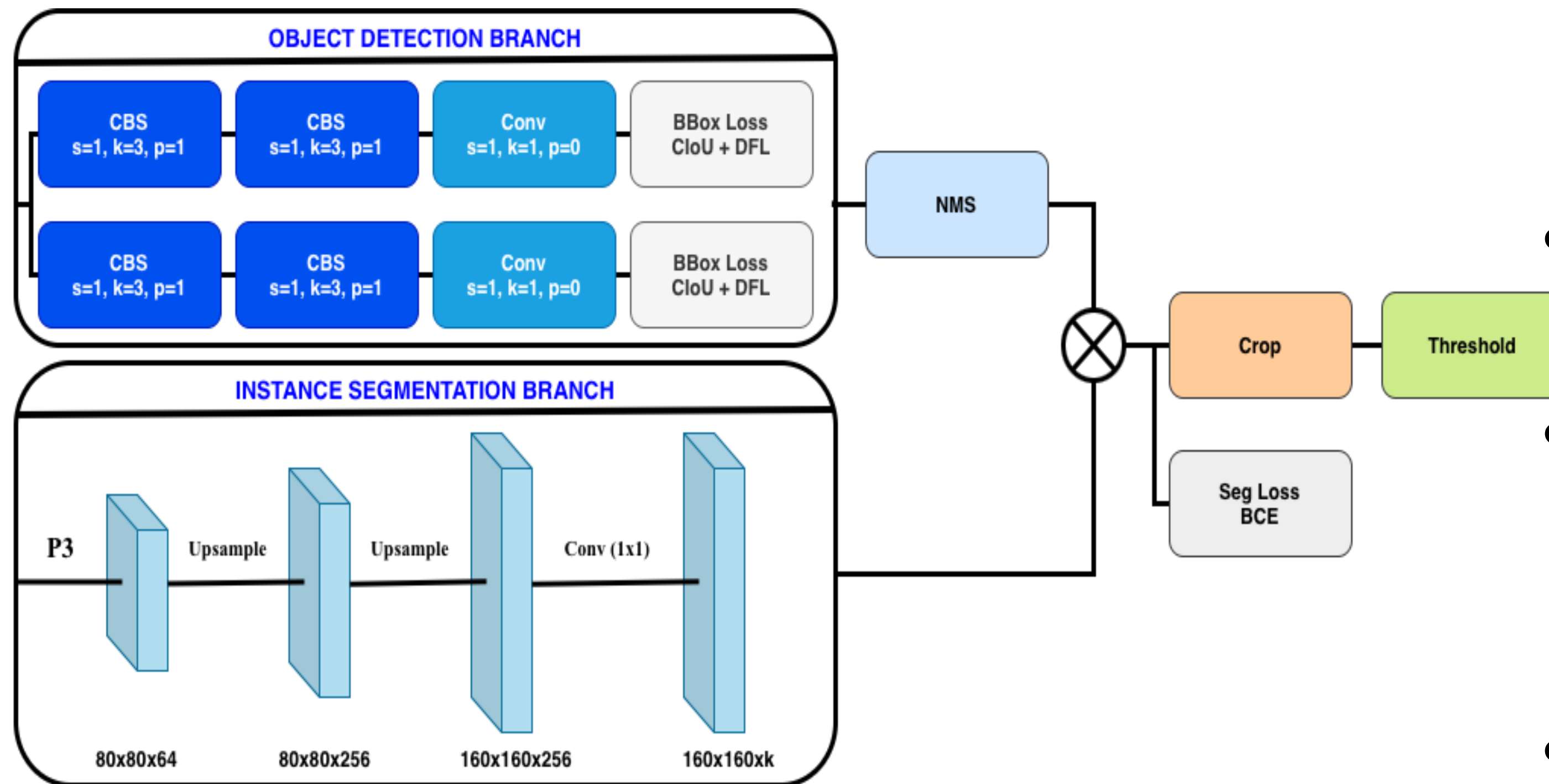
$y[i]$ - feature map, K - weights count, k - current weight, w - convolution kernel

- ASPP Module

$$Y_{ASPP} = Conv_{1 \times 1}([Y_{1 \times 1}, Y_{r=6}, Y_{r=12}, Y_{r=18}, Y_{BilinearUpSample}])$$

r - resolution, $[...]$ - concatenation, $Y_{BilinearUpSample}$ - upsampling

Combined neural networks



YOLO-seg architecture (at level P3)

- YOLACT Principle
- Encoder CSPDarknet

$$P_3, P_4, P_5 = \frac{H}{8} \times \frac{W}{8}, \frac{H}{16} \times \frac{W}{16}, \frac{H}{32} \times \frac{W}{32}$$

- Generation of masks instances (neck)

$$P = ProtoNet(P_{3,4,5}) \in \mathbb{R}^{k \times \frac{H}{8} \times \frac{W}{8} \dots \frac{H}{32} \times \frac{W}{32}}$$

- Class prediction (head)

$$C_i = Detect_{bbox}(H_{bbox}, W_{bbox}, P_{bbox})$$

$$SegClass = \operatorname{argmax} [C_1, C_2, \dots, C_N]$$

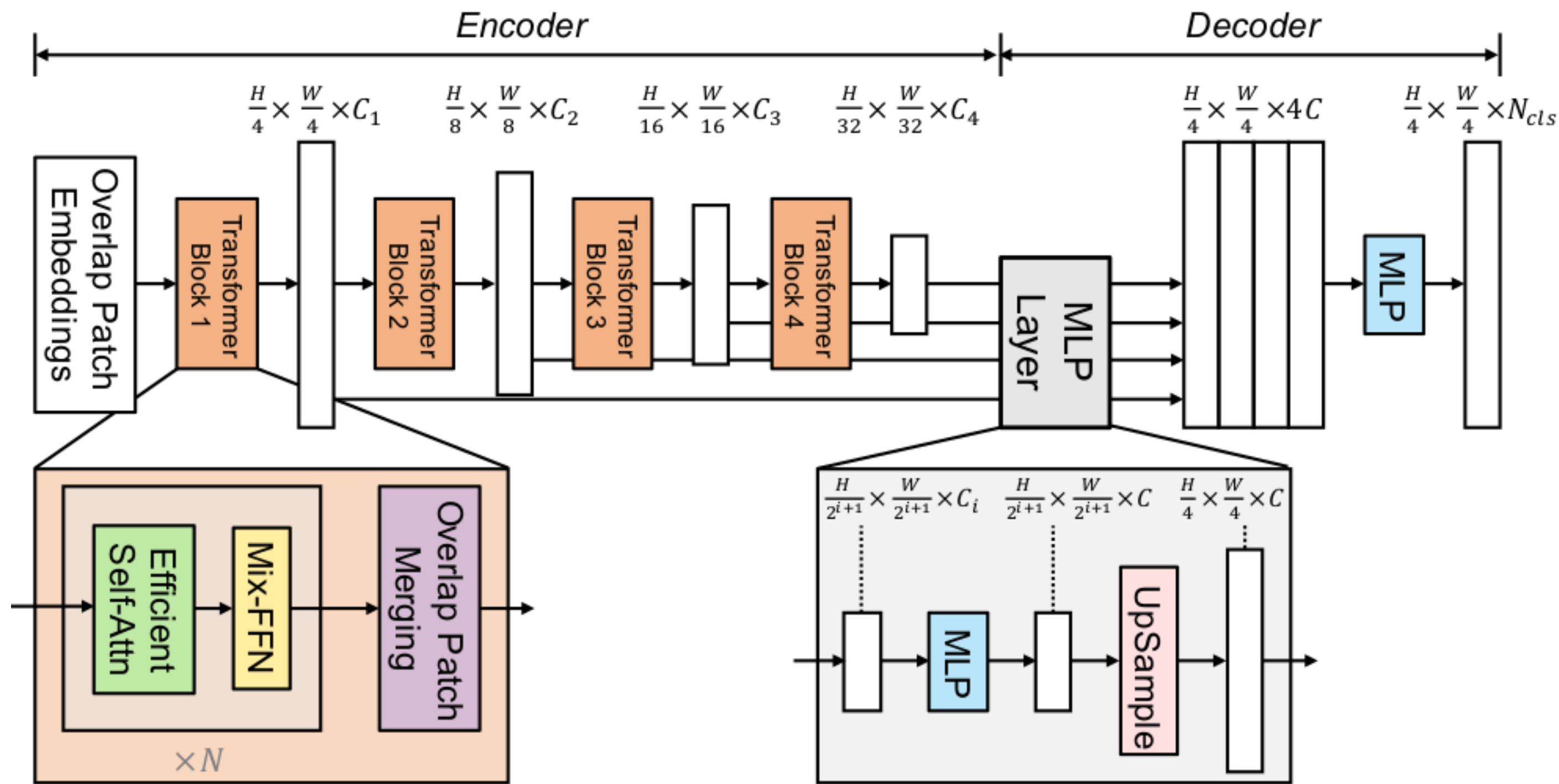
$$ProtoMask : [32 \times 32], Coef : \parallel 32 \parallel$$

- Mask generation

$$M_i = \sigma \left(\sum_{j=1}^k a_{i,j} \cdot P_j \right) = \sigma(A_i \cdot P)$$

M_i - mask, i - detection index, $a_{i,j}$ - weights, P_j - tensor of instances (pixels)

Visual transformers



SegFormer architecture

- Patch splitting

$$F_i = \text{LinearProjection}(\text{PatchMerge}(F_{i-1}))$$

F_i - tensor, PatchMerge - patch of pixels, LinearProjection - reshape (projection)

- Effective self attention (ESA)

$$N_{reduced} = \frac{H}{R} \times \frac{W}{R} \quad \hat{K} = \text{Reshape}\left(\frac{N_{reduced}}{R} \times (R \cdot C), K\right)$$

$$K_{new} = \text{LinearProjection}(\text{Conv}_{1 \times 1, \text{stride}=R} \hat{K})$$

$$\text{Attention}(Q, K_{new}, V_{new}) = \text{Softmax}\left(\frac{QK_{new}^T}{\sqrt{d_{head}}}\right)V_{new}$$

Q - query matrix $[N, d]$, K_{new} - key matrix, V_{new} - values matrix, d_{head} - attention head

- Decoder: MLP

Upsampling and concatenation

RailSem19 Dataset



RailSem19

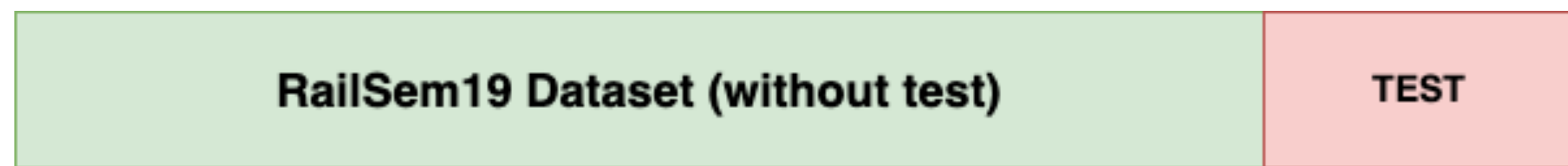
```
{
  "frame":
  "rs00000",
  "imgHeight":
  1080,
  "imgWidth": 1920,
  "objects": [
    {
      "boundingbox": [
        1066,
        571,
        1077,
        590
      ],
      "label":
      "track-sign-front"
    },
  ],
}
```

→ $\langle class \rangle \langle x_1 \rangle \langle y_1 \rangle , \dots , \langle x_n \rangle \langle y_n \rangle$

Example of notation JSON (left) and YOLO (right)

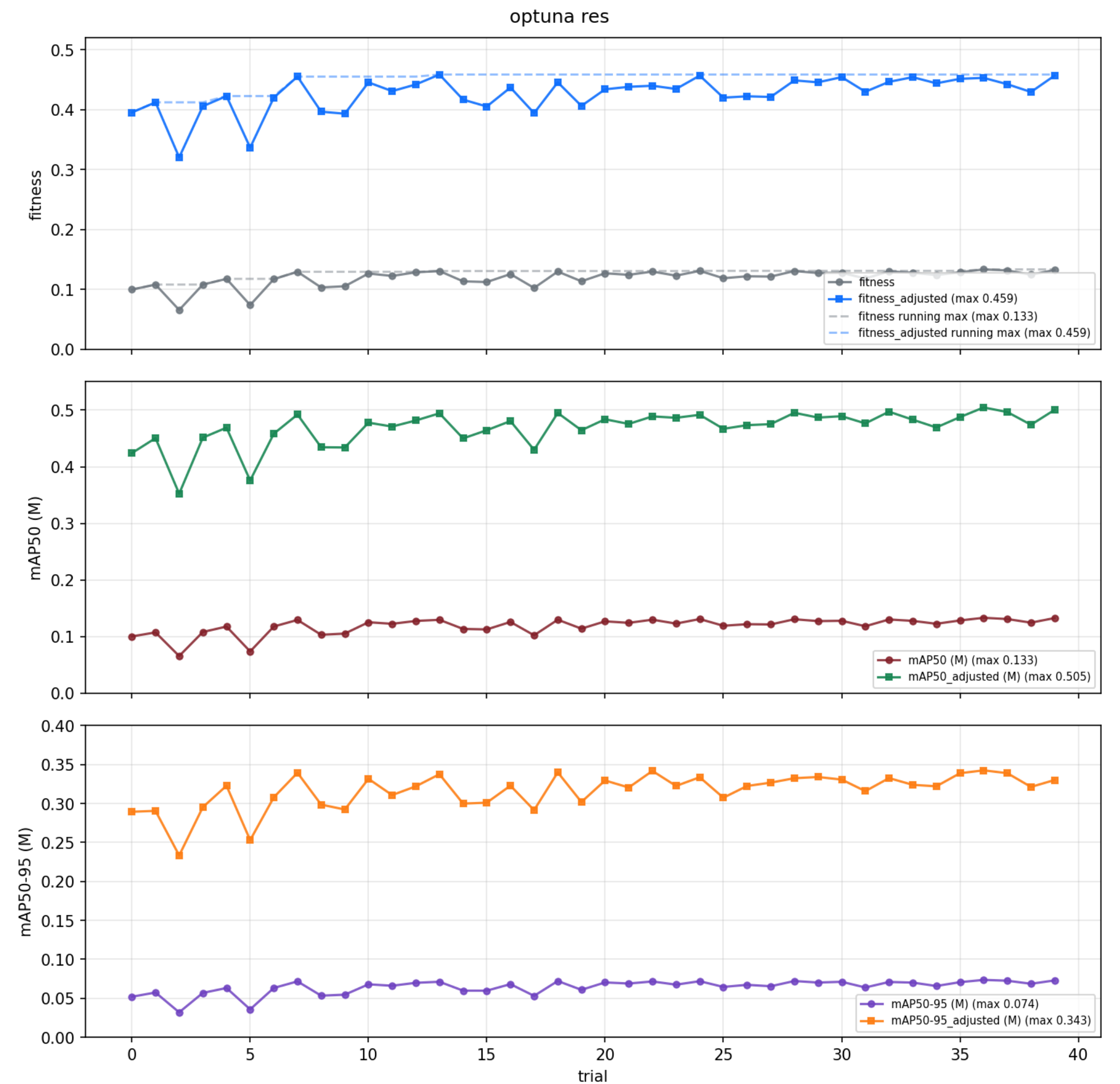
Preparing to train

- TPE based optuna, 40/30;
- Cross validation ($K = 4$).



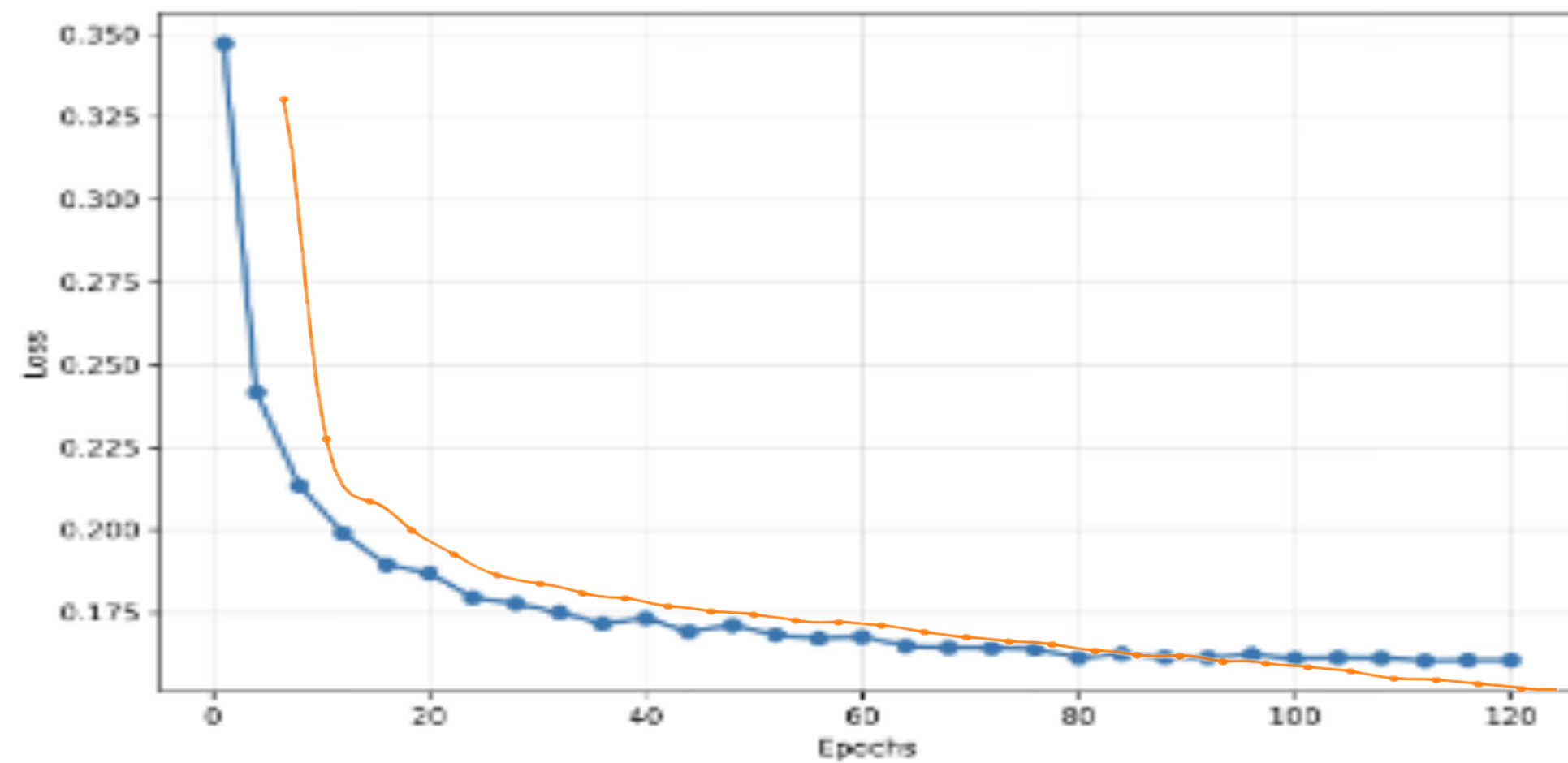
TRAIN	TRAIN	TRAIN	VAL
TRAIN	TRAIN	VAL	TRAIN
TRAIN	VAL	TRAIN	TRAIN
VAL	TRAIN	TRAIN	TRAIN

Cross validation with 4 folds

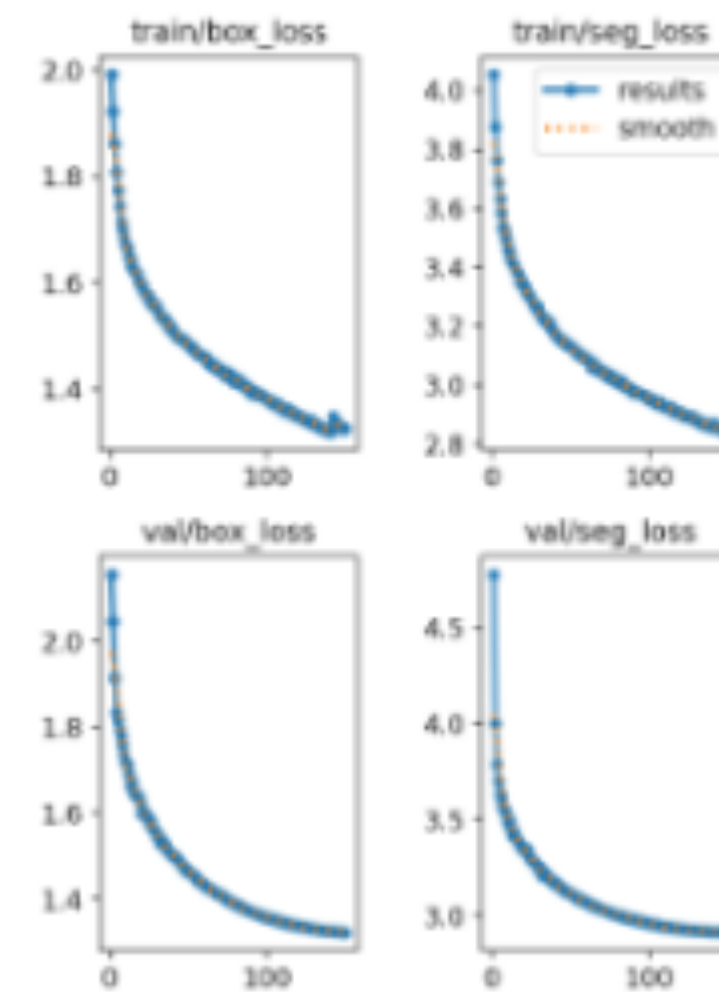


Best hyperparameters

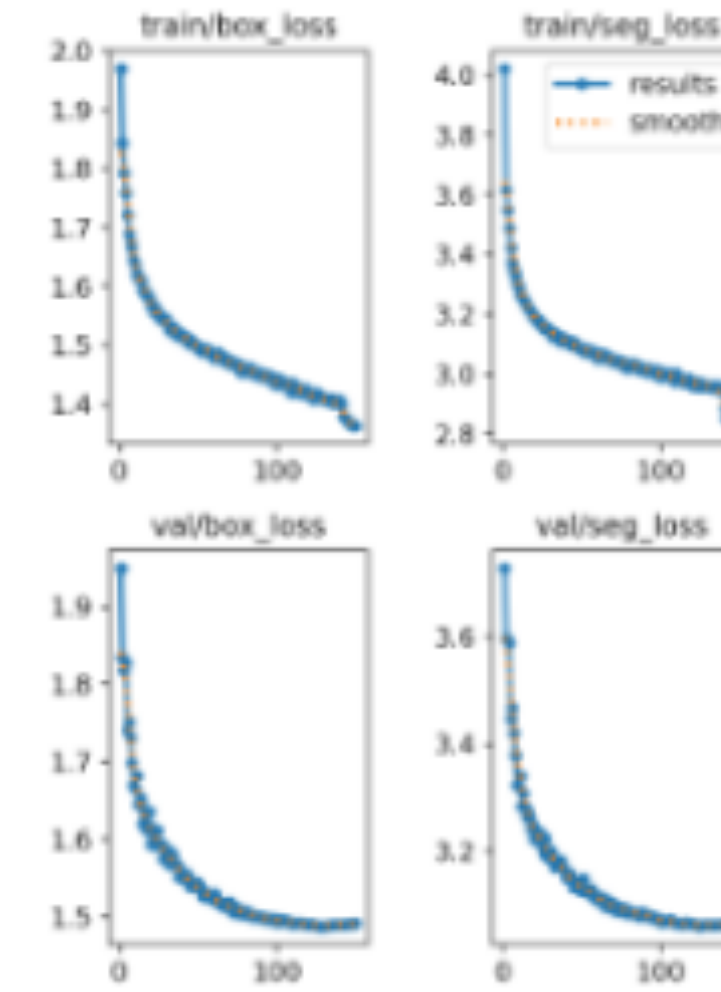
Train results



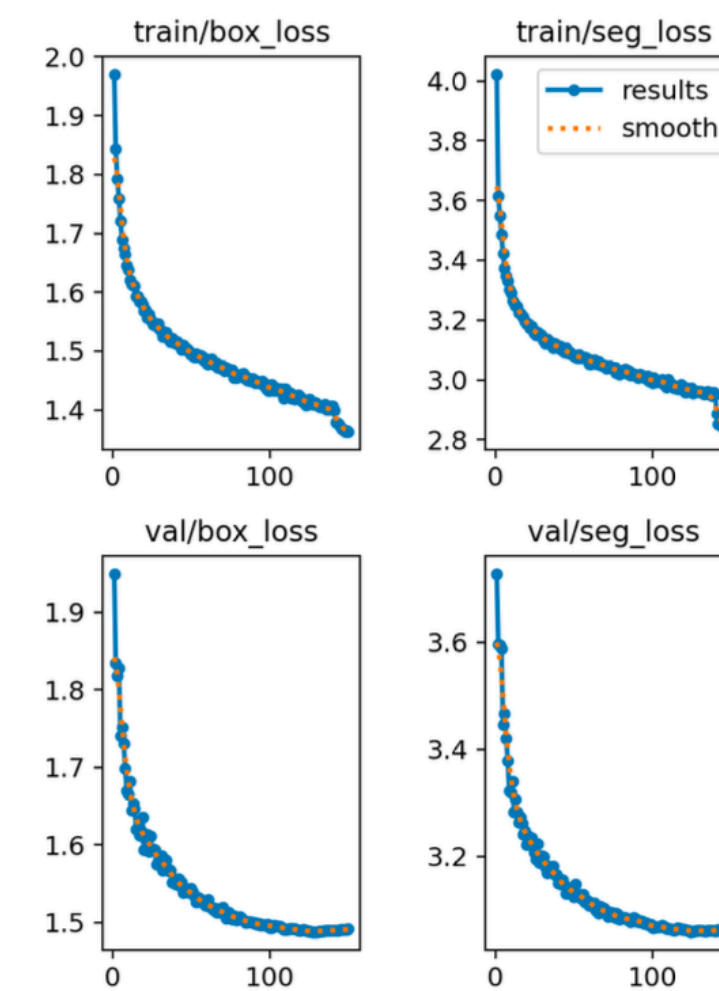
SegFormer (B0)



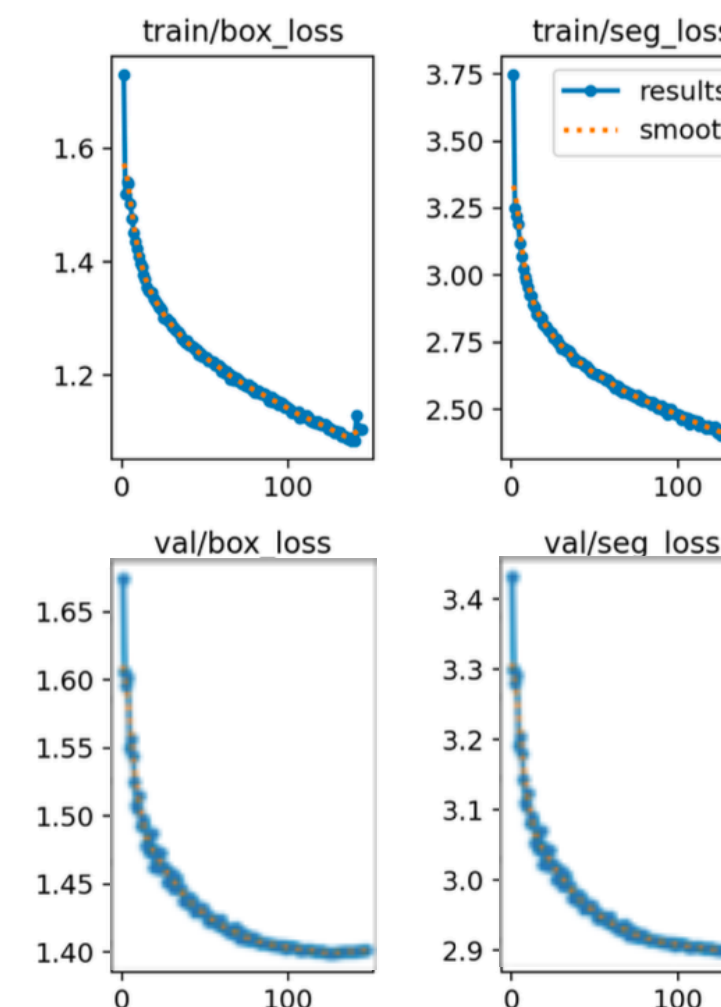
YOLOv8n-seg



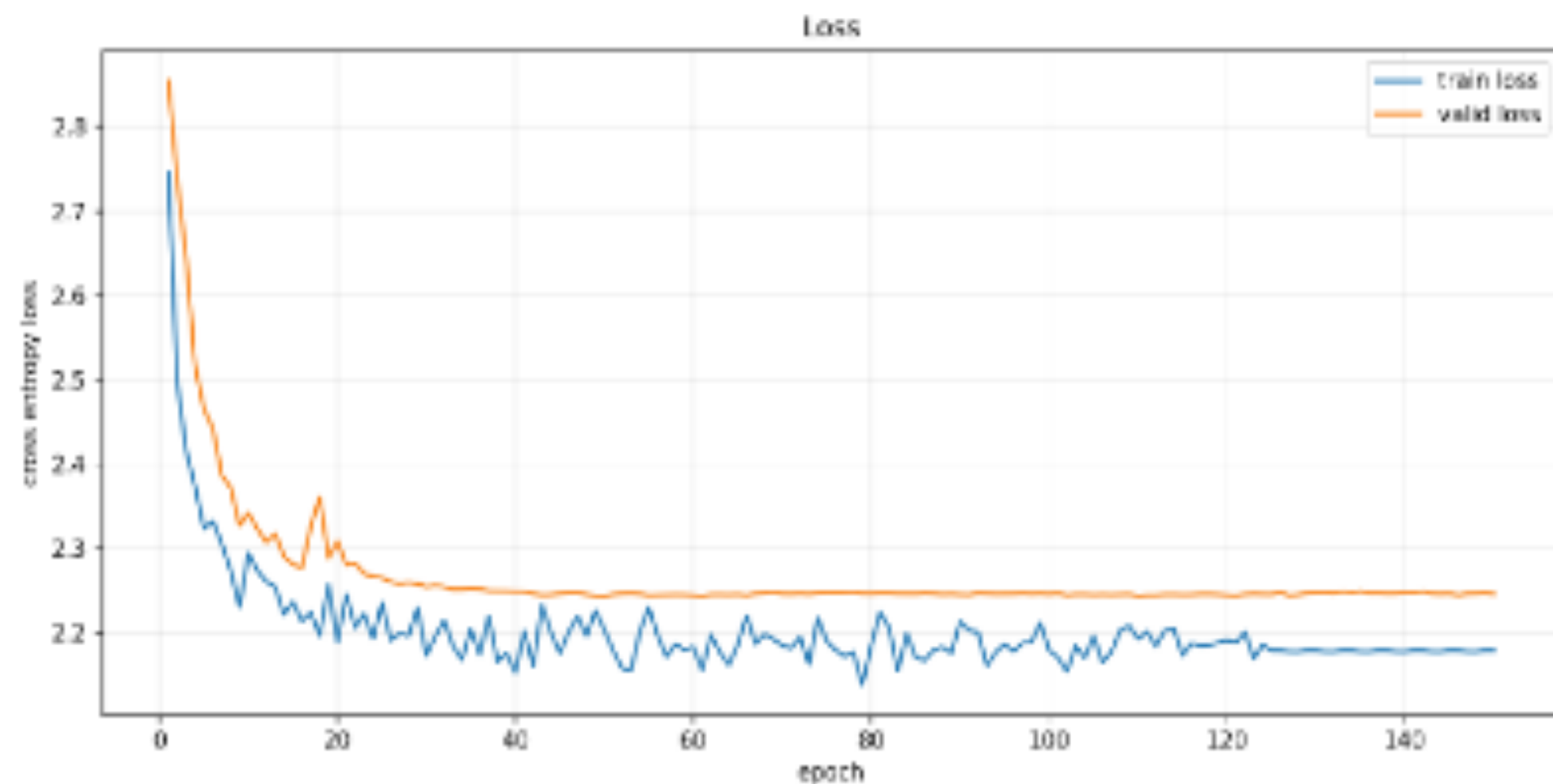
YOLOv11s-seg



YOLOv8s-seg



YOLO26s-seg

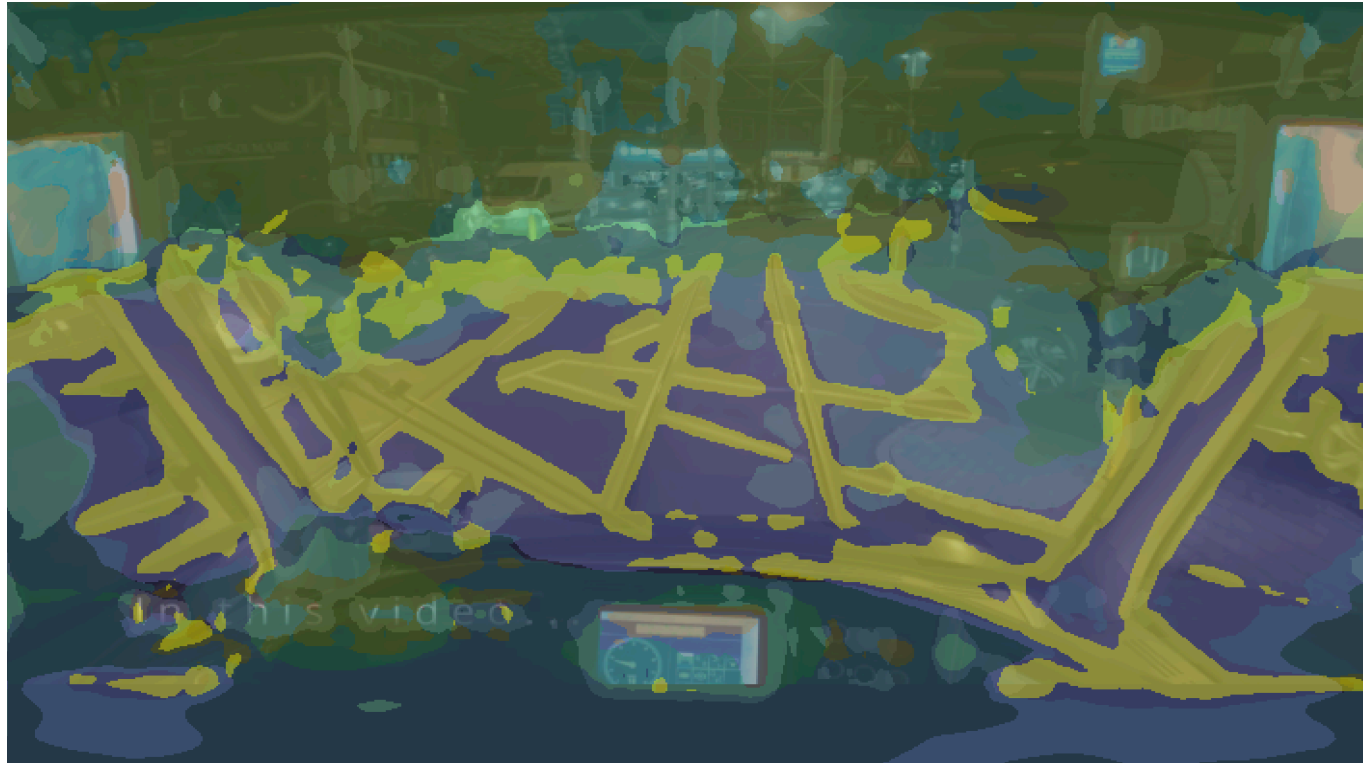


DeepLabV3+(ResNet50)

Results on test

Models / Quality	DeepLabV3+	YOLO-seg	SegFormer
mIoU	64%	81%	71% (on B0!)
mDice	78%	89%	83%
FPS	< 1	56	32

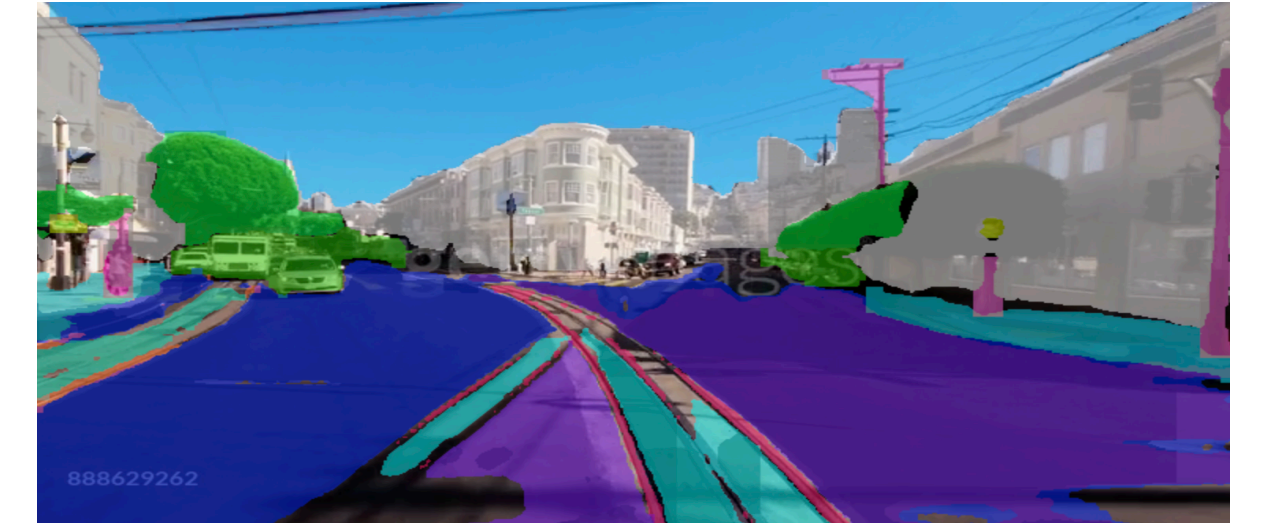
Examples



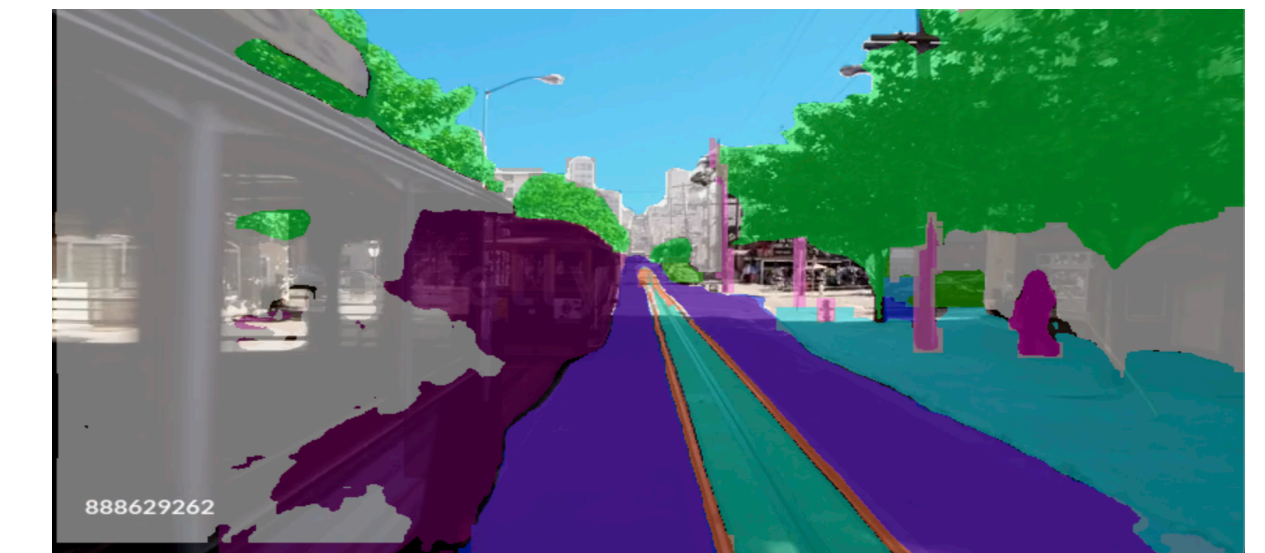
DeepLabV3+ (ResNet50)



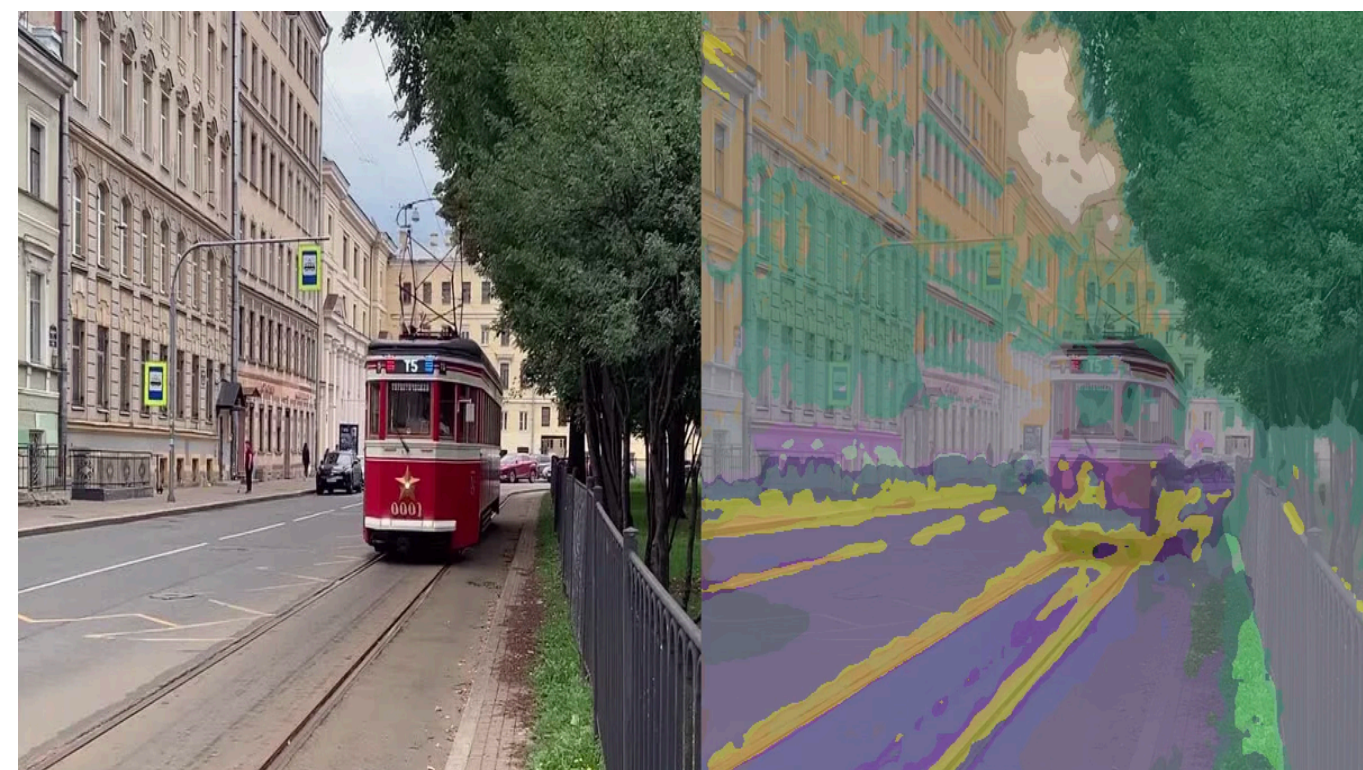
SegFormer (B0)



YOLO-seg (8n)



YOLO-seg (11s)



DeepLabV3+ (ResNet101)



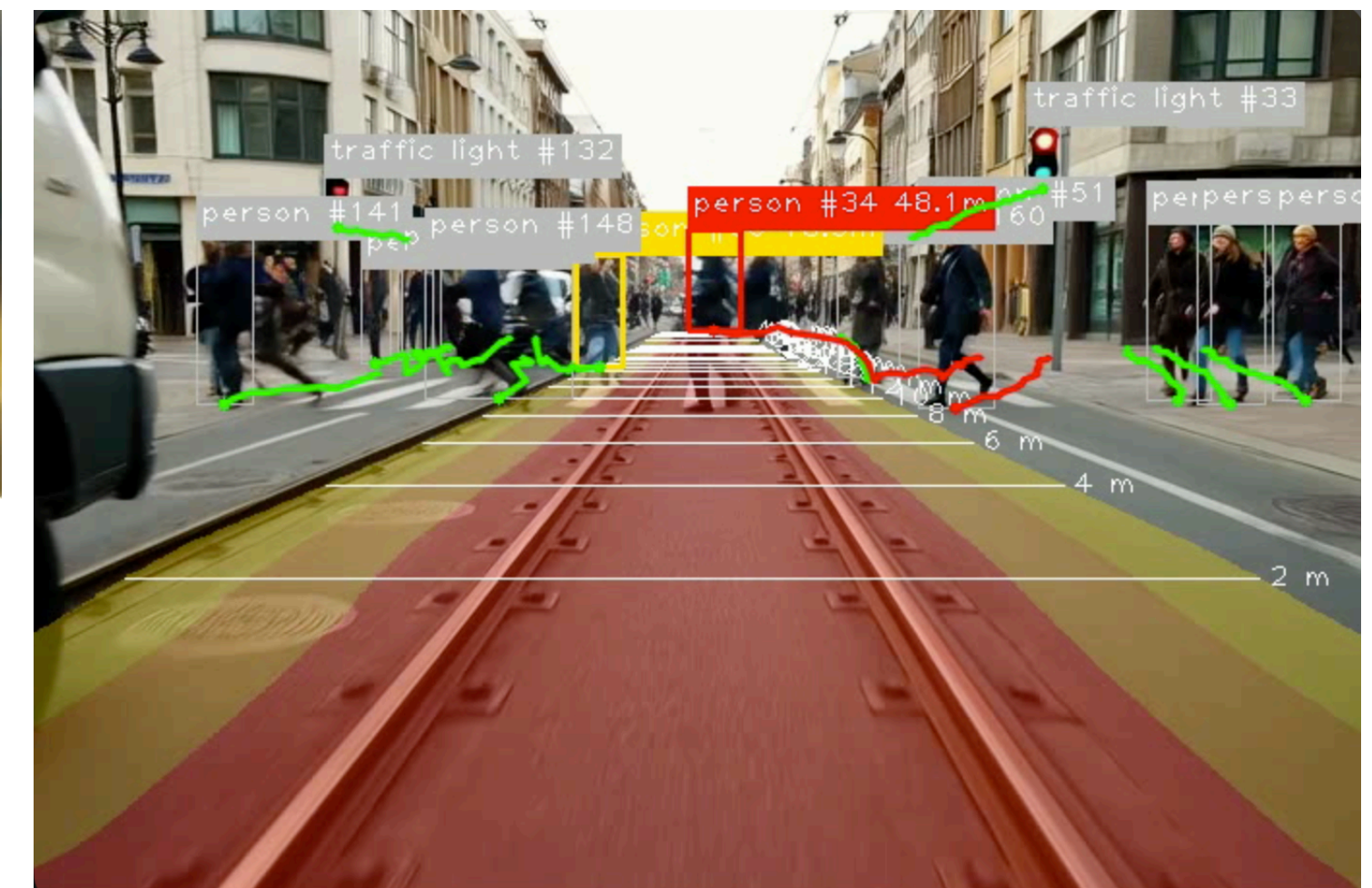
YOLO-seg (26s)



YOLO-seg (8s)

Conclusion

- **Answer: No... Each task requires its own model**
 - Real-time segmentation
 - YOLO-seg (FPS: 56, mIoU: 81%, mDice: 89%)
 - Large objects, static segmentation and fast learning
 - DeepLabV3+ (FPS: < 1, mIoU: 64%, mDice: 78%)
 - Tiny and small objects static segmentation
 - SegFormer (FPS: 32, mIoU: 71%, mDice: 83%)
 - Results of algorithmic and tracking part
 - Added tracker ByteTrack with Kalman filter
 - Ego-Track logic
 - Detection (from 3 detectors chosen YOLOv8n)
 - Added dynamic RoI



Final results

Thanks for your attention!

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